Assignment 2:Dynamic Memory allocation in C

Q1.Write c program to reverse an array elements using Dynamic Memory Allocation

```
#include <stdio.h>
#include<stdlib.h>
int main()
{
int *ip,n,i;
int max=0;
printf("\nEnter the limit: ");
scanf("%d",&n);
ip=(int*)malloc(n*sizeof(int));
printf("\nEnter the elements:\n");
for(i=0;i<n;i++)
{
scanf("%d",&ip[i]);
}
printf("\nEntered elements are:\n");
for(i=0;i<n;i++)
{
printf("\n%d",ip[i]);
printf("\n");
}
printf("\nReverse array is:");
for(i=n-1;i>=0;i--)
{
printf("\n%d",ip[i]);
}
return 0;
}
```

Q2.Write a C Program to demonstrate malloc() & free() Function.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
   int i, n;
   int *ptr;
   printf("Enter total number of elements: ");
   scanf("%d", &n);
   ptr = (int *) malloc(n * sizeof(int));
   if(ptr == NULL)
   {
      printf("Memory is not created!!!");
      exit(0);
   }
   else
   {
    printf("Memory  created Successfully!!!");
   printf("Enter %d elements: \n", n);
   for (i = 0; i < n; i++)
      scanf("%d", (ptr + i));
     printf("\nArray elements are:\n");
   for (i = 0; i < n; i++)
      printf("%d ", *(ptr + i));

      free(ptr);
   }
   return 0;
}
```

Q3.Write a C Program to demonstrate calloc() & free() Function.

```
#include <stdio.h>
```

```
#include <stdlib.h>

int main()
{
    int i,n;
    int *ptr;
    printf("Enter total number of elements: ");
    scanf("%d", &n);
    ptr = (int *) calloc(n, sizeof(int));
    if(ptr == NULL)
    {
        printf("Memory is not created!!!");
        exit(0);
    }
    printf("Enter %d elements: \n", n);
    for (i = 0; i <n; i++)
        scanf("%d", (ptr + i));

    printf("\nArray elements are:\n");
    for (i = 0; i < n; i++)
        printf("%d ", *(ptr + i));

    free(ptr);

    return 0;
}
```

Q4. C program to find sum of array elements using Dynamic Memory Allocation

```
#include <stdio.h>
#include <stdlib.h>
int main()
```

```
{
   int* ptr;
   int limit;
   int i;
   int sum;
   printf("Enter limit of the array: ");
   scanf("%d", &limit);
   ptr = (int*)malloc(limit * sizeof(int));
   for (i = 0; i < limit; i++)
 {
      printf("Enter element %02d: ", i + 1);
      scanf("%d", (ptr + i));
   }
   printf("\nEntered array elements are:\n");
   for (i = 0; i < limit; i++) {
      printf("%d\n", *(ptr + i));
   }
   sum = 0; //assign 0 to replace garbage value
   for (i = 0; i < limit; i++) {
      sum += *(ptr + i);
   }
   printf("Sum of array elements is: %d\n", sum);
   free(ptr);
   return 0;
}
```

Q5.Accept N integer in an array using dynamic memory allocation.Find maximum from them & display.

```
#include <stdio.h>
#include <stdlib.h>
int main()
 {
  int n;
```

```
  int *data;
  printf("Enter the total number of elements: ");
  scanf("%d", &n);
  data = (int *)calloc(n, sizeof(int));
  if (data == NULL)
{
    printf("Error!!! memory not allocated.");
    exit(0);
  }
  for (int i = 0; i < n; ++i)
  {
    printf("Enter number%d: ", i + 1);
    scanf("%d", data + i);
  }
  for (int i = 1; i < n; ++i)
{
    if (*data < *(data + i))
 {
      *data = *(data + i);
    }
  }
  printf("Largest number = %d", *data);

  free(data);

  return 0;
}
```

Q6. Accept n integers in an array. Copy only the non-zero elements to another array (allocated using dynamic memory allocation). Calculate the sum and average of non-zero elements.

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int a[30], i,n, sum = 0;
    int count=0;
    int *ptr;
    float avg;
    printf("\n Enter the total number of elements you want to enter : ");
    scanf("%d",&n);
    printf("\n Enter element in aaray");
        for(i = 0;i<n;i++)
        {
        scanf("%d",&a[i]);
        }
       for(i = 0;i<n;i++)
        {
            if(a[i]!=0)
            {
            count++;
            }
    }
    ptr = (int *)malloc(count * sizeof(int));
    for(i = 0;i<n;i++)
    {
            if(a[i]!=0)
            {
         *(ptr+i)=a[i];
                sum=sum+*(ptr+i);
                }
                }
    avg=sum/count;
    printf("sum is %d \n",sum);
    printf("avg is %f\n",avg);
        return 0;
  }
```

Q7. write a c program accept a matrix of order MxN (use dynamic memory allocation & array of pointers concept)Display trace of the matrix.

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
	int m, n;
	printf("Enter no. of rows and columns: ");
	scanf("%d%d", &m, &n);
	int **a;
	//Allocate memory to matrix
	a = (int **) malloc(m * sizeof(int *));
	for(int i=0; i<m; i++)
	{
		a[i] = (int *) malloc(n * sizeof(int));
	}
	//Read elements into matrix
	printf("Enter matrix elements: ");
	for(int i=0; i<m; i++)
	{
		for(int j=0; j<n; j++)
		{
			scanf("%d", &a[i][j]);
		}
	}
	//Print elements in the matrix
	printf("Matrix elements are: \n");
	for(int i=0; i<m; i++)
	{
		for(int j=0; j<n; j++)
		{
			printf("%d ", a[i][j]);
		}
		printf("\n");
	}
	//Dellocating memory of matrix
	for(int i=0; i<m; i++)
	{
		free(a[i]);
	}
	free(a);
	return 0;
}
```